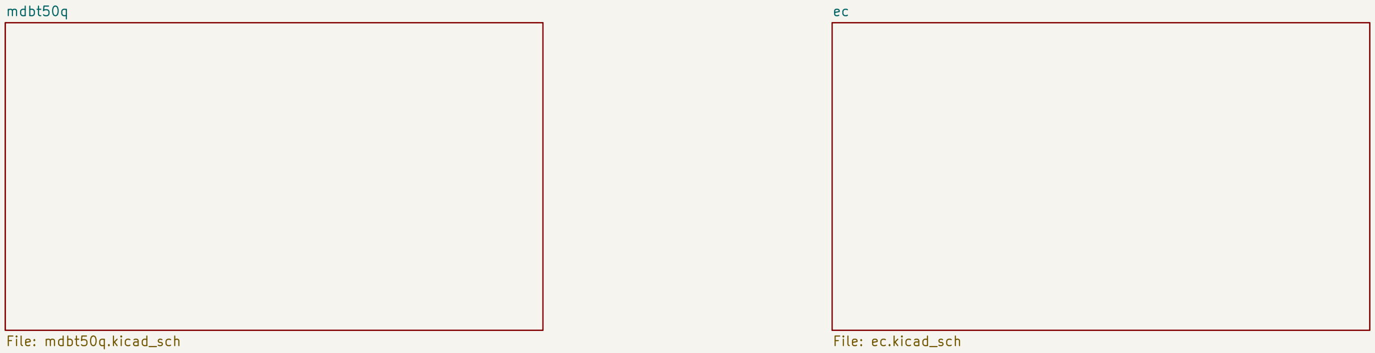
TC52 wireless: MDBT50Q-1MV2 (nRF52840) + 1S LiPo version of the wired TC52 capacitive keyboard.
Sheet mdbt50q: module, charger, USB, SWD, reset, encoder.  Sheet ec: 52 electrodes, MUX x2, analog front end.
The sheets are joined by global labels: ROW1..4_DRV, INHA/S1A..S3A, INHB/S1B..S3B, SENSE (purge, P1.02),
ADC (AIN4 = P0.28), AFE_EN (P0.13), ENC_A / ENC_B / ENC_SW, +3V3 (module VDD = REG0 output), GND.
Pin table and firmware notes: docs/tc52w-build-log.md



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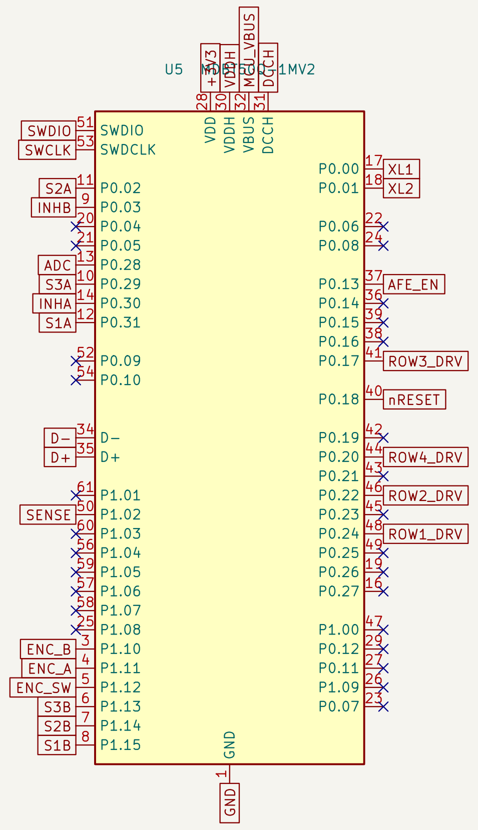
Sheet: /
File: ts52k.kicad_sch

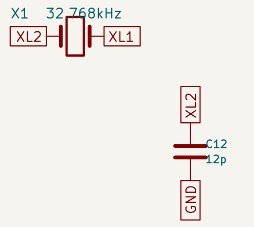
**Title: TC52 wireless**

Size: A4
Date:
**Rev: 0.1**
KiCad E.D.A. 10.0.8
Id: 1/3

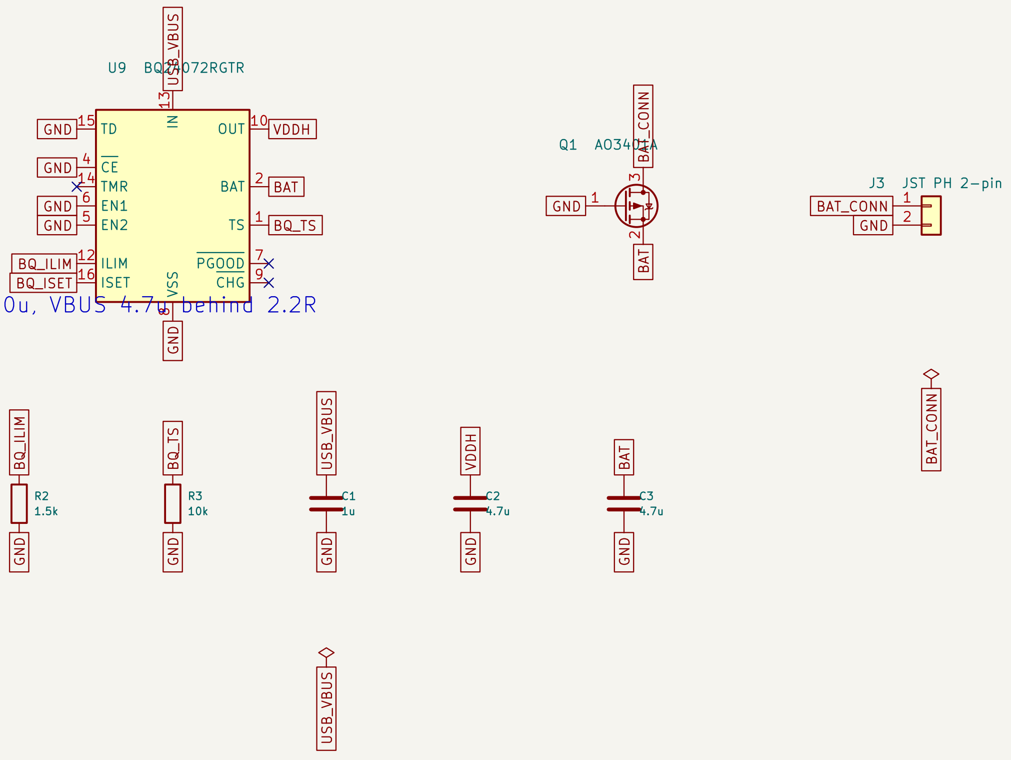
MDBT50Q−1MV2 (nRF52840). Pads 3..14 are 'standard drive, low frequency I/O only' (Raytac spec 2.6): they carry the encoder and the MUX select / inhibit lines (change once per column, well under 10 kHz) and the ADC input (AIN4).
Row strobes are on unrestricted pads; PURGE (P1.02) only ever sees the millivolt−level SENSE node. P0.15 (boot loader LED), NFC pins and P0.26/P0.27 (next to XL1/XL2) are unused.


U5 MDBT50Q-1MV2
+3V3
VDDH
MCU_VBUS
DCCH
VDD
VDDH
VBUS
DCCH
SWDIO
SWCLK
SWDCLK
S2A
INHB
ADC
S3A
INHA
S1A
D−
D+
SENSE
ENC_B
ENC_A
ENC_SW
S3B
S2B
S1B
P0.02
P0.03
P0.04
P0.05
P0.28
P0.29
P0.30
P0.31
P0.09
P0.10
P1.01
P1.02
P1.03
P1.04
P1.05
P1.06
P1.07
P1.08
P1.10
P1.11
P1.12
P1.13
P1.14
P1.15
P0.00
P0.01
P0.06
P0.08
P0.13
P0.14
P0.15
P0.16
P0.17
P0.18
P0.19
P0.20
P0.21
P0.22
P0.23
P0.24
P0.25
P0.26
P0.27
P1.00
P0.12
P0.11
P1.09
P0.07
XL1
XL2
AFE_EN
ROW3_DRV
nRESET
ROW4_DRV
ROW2_DRV
ROW1_DRV
GND

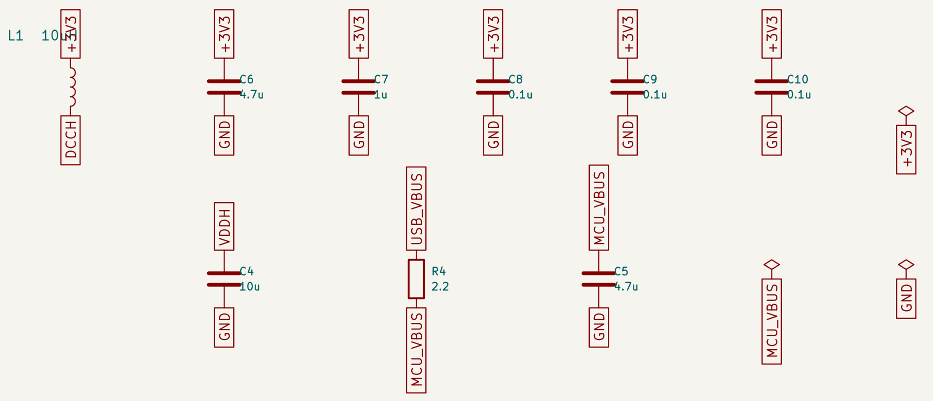
32.768 kHz crystal (CL 9 pF), 12 pF x2


X1 32.768kHz
XL2
XL1
C11 12p
C12 12p
GND

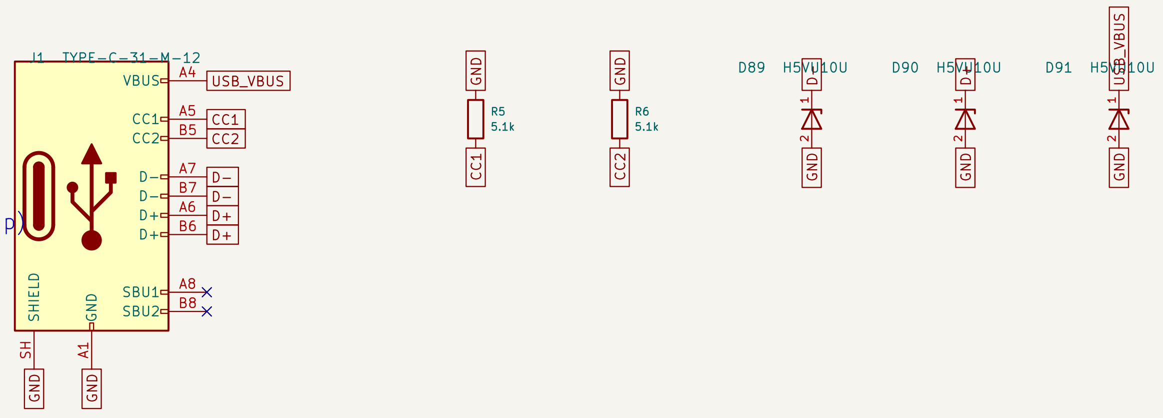
BQ24072: USB100 fixed (EN1 = EN2 = CE = TD = GND), ISET 8.2k, ILIM 1.5k, TS 10k, TMR open. OUT −> VDDH.
Q1 = battery reverse−polarity protection (drain to the connector, source to BAT, gate to GND). J3: 1 = BAT+, 2 = GND


U9 BQ24072RGTR
USB_VBUS
IN
TD
CE
TMR
EN1
EN2
ILIM
ISET
OUT
BAT
TS
PGOOD
CHG
VSS
VDDH
BQ_TS
BQ_ILIM
BQ_ISET
GND
Q1 AO3401A
BAT_CONN
BAT
J3 JST PH 2-pin
R1 8.2k
R2 1.5k
R3 10k
C1 1u
C2 4.7u
C3 4.7u

Module supply (TS52K wireless): REG0 DC/DC inductor DCCH − VDD, VDD 4.7u + 1u + 100n x3, VDDH 10u, VBUS 4.7u behind 2.2R

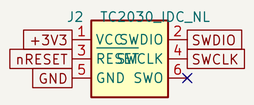

L1 10uH
+3V3
DCCH
C6 4.7u
C7 1u
C8 0.1u
C9 0.1u
C10 0.1u
C4 10u
VDDH
USB_VBUS
R4 2.2
MCU_VBUS
C5 4.7u
GND

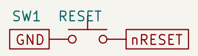
USB−C (USB 2.0 full speed, no series resistors), CC 5.1k x2, ESD H5VU10U x3 at the connector

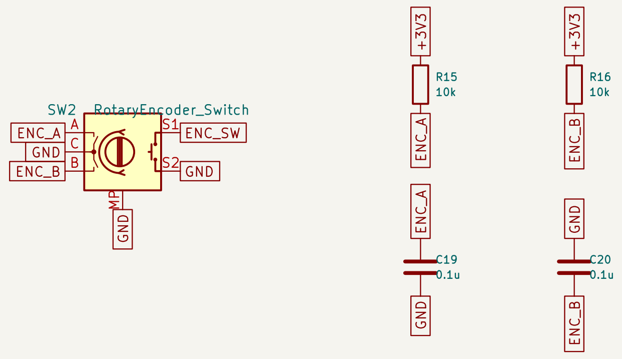

J1 TYPE-C-31-M-12
VBUS
USB_VBUS
CC1
CC2
D−
D+
SBU1
SBU2
SHIELD
GND
R5 5.1k
R6 5.1k
D89 H5VU10U
D90 H5VU10U
D91 H5VU10U

SWD contacts (Tag−Connect TC2030−NL, no part fitted): 1 VTref = VDD, 2 SWDIO, 3 nRESET, 4 SWCLK, 5 GND. SW1 = reset (P0.18)

Encoder (as wired TC52): A/B with 10k pull−ups and 100n, push switch to GND (internal pull−up)


J2 TC2030_IDC_NL
+3V3
nRESET
GND
VCC
RESET
SWDIO
SWCLK
SWO


SW1 RESET
GND
nRESET


SW2 RotaryEncoder_Switch
ENC_A
GND
ENC_B
ENC_SW
R15 10k
R16 10k
+3V3
C19 0.1u
C20 0.1u

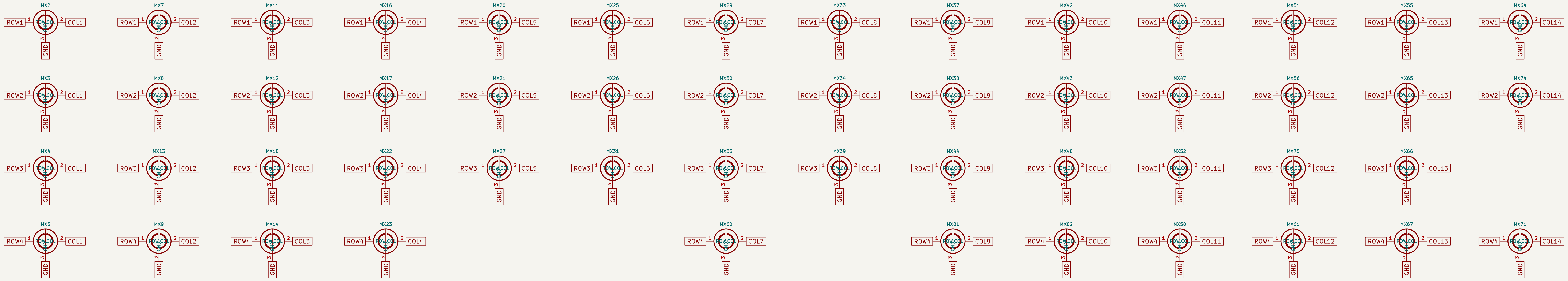
Generated by pcb/tc52w−build/gen_sch.py

Sheet: /mdbt50q/
File: mdbt50q.kicad_sch

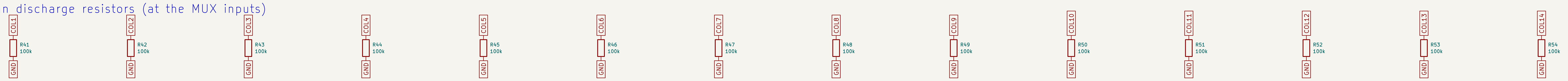
**Title: TC52 wireless: MDBT50Q, power, USB, SWD, encoder**

Size: A2 Date: **Rev: 0.1**
KiCad E.D.A. 10.0.8 Id: 2/3

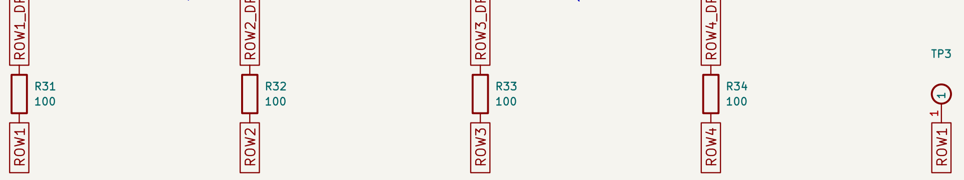
Key electrodes: 14 columns x 4 rows (design−spec 3). Pad 1 = drive (ROW), pad 2 = sense (COL), pad 3 = guard ring


MX2
MX7
MX11
MX16
MX20
MX25
MX29
MX33
MX37
MX42
MX46
MX51
MX55
MX64
MX3
MX8
MX12
MX17
MX21
MX26
MX30
MX34
MX38
MX43
MX47
MX56
MX65
MX74
MX4
MX13
MX18
MX22
MX27
MX31
MX35
MX39
MX44
MX48
MX52
MX75
MX66
MX5
MX9
MX14
MX23
MX60
MX81
MX82
MX58
MX61
MX67
MX71
ROW1
ROW2
ROW3
ROW4
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
COL9
COL10
COL11
COL12
COL13
COL14
ROW COL
GND
1
2
3

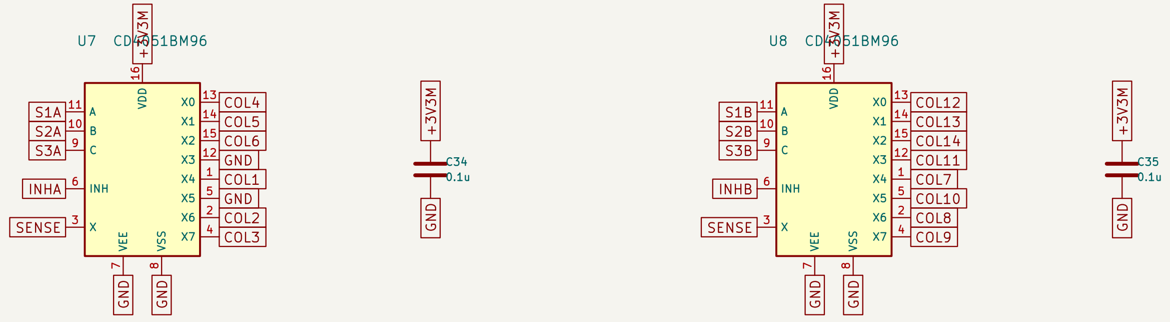
Column discharge resistors (at the MUX inputs)


COL1
R41
100k
COL2
R42
100k
COL3
R43
100k
COL4
R44
100k
COL5
R45
100k
COL6
R46
100k
COL7
R47
100k
COL8
R48
100k
COL9
R49
100k
COL10
R50
100k
COL11
R51
100k
COL12
R52
100k
COL13
R53
100k
COL14
R54
100k
GND

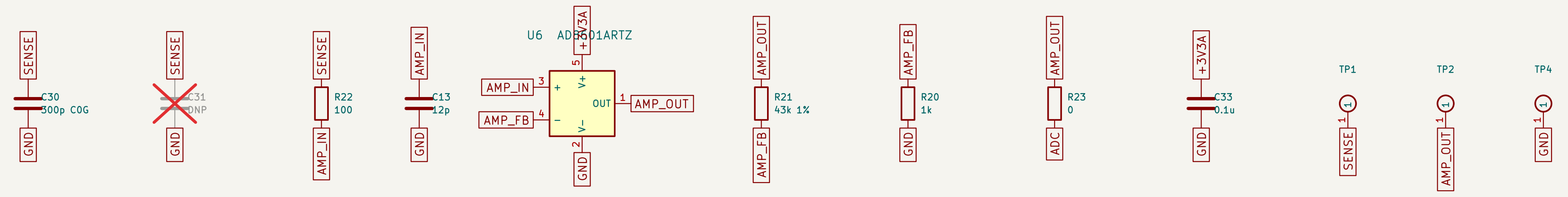
Row drive: module pads 48 / 46 / 41 / 44 (P0.24, P0.22, P0.17, P0.20) −> 100R −> ROW1..4


ROW1_DRV
R31
100
ROW1
ROW2_DRV
R32
100
ROW2
ROW3_DRV
R33
100
ROW3
ROW4_DRV
R34
100
ROW4
TP3
1
ROW1

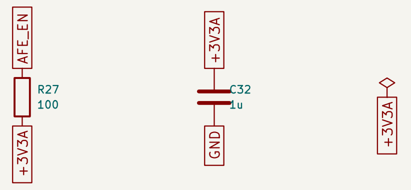
Column select: each MUX has its own lines. INHA / INHB low = that MUX on (never both low). S1x/S2x/S3x = address A/B/C


U7 CD4051BM96
+3V3M
VDD
S1A
S2A
S3A
A
B
C
INHA
INH
SENSE
X
VEE
VSS
X0
X1
X2
X3
X4
X5
X6
X7
COL4
COL5
COL6
GND
COL1
GND
COL2
COL3
GND
GND
C34
0.1u
+3V3M
GND
U8 CD4051BM96
+3V3M
S1B
S2B
S3B
INHB
SENSE
COL12
COL13
COL14
COL11
COL7
COL10
COL8
COL9
GND
GND
C35
0.1u
+3V3M
GND

Analog front end (wired TC52 circuit): SENSE node (C30 reference cap, purge by P1.02) −> R22 −> non−inverting x44 −> AIN4 (P0.28).
C13 (12 pF at the + input, with R22: 130 MHz low−pass) keeps 2.4 GHz from the antenna out of the op amp input


SENSE
C30
300p C0G
GND
SENSE
C31
DNP
GND
SENSE
R22
100
AMP_IN
AMP_IN
C13
12p
GND
U6 AD8601ARTZ
+3V3A
AMP_IN
AMP_FB
+
−
V+
V−
OUT
AMP_OUT
GND
AMP_OUT
R21
43k 1%
AMP_FB
AMP_FB
R20
1k
GND
AMP_OUT
R23
0
ADC
+3V3A
C33
0.1u
GND
TP1
SENSE
TP2
AMP_OUT
TP4
GND

+3V3A (op amp) = GPIO AFE_EN (P0.13, high drive, held high while scanning) through R27 100R, 1u + 100n.
The op amp draws under 1 mA; the pin also gates it for sleep. 100R / 1u (not 22R / 4.7u as on +3V3M): the pin charges the capacitors at switch−on, this keeps the peak near 25 mA and the charge at 3.6 uC (same 1.4 kHz corner).
Reason for not feeding it from +3V3: docs/tc52w−build−log.md


AFE_EN
R27
100
+3V3A
+3V3A
C32
1u
GND
+3V3A

+3V3M (MUX x2) = +3V3 through 22R, 4.7u

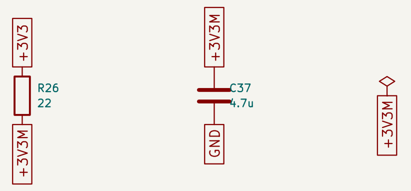
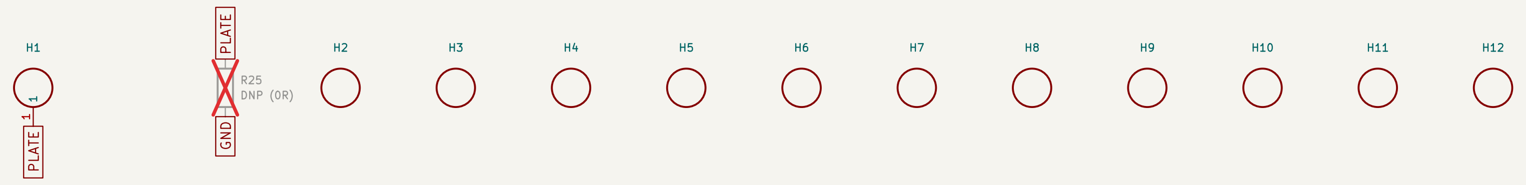

+3V3
R26
22
+3V3M
+3V3M
C37
4.7u
GND
+3V3M

Plate fixing holes (M2). H1 has a land: PLATE −> R25 (not fitted) −> GND


H1
1
PLATE
PLATE
R25
DNP (0R)
GND
H2
H3
H4
H5
H6
H7
H8
H9
H10
H11
H12

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Sheet: /ec/
File: ec.kicad_sch
**Title: TC52 wireless: capacitive matrix and analog front end**
Size: A2
Date:
**Rev: 0.1**
KiCad E.D.A. 10.0.8
Id: 3/3